FINAL EXPLANATION: PHYSICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: PHYSICS GRADE 10 — SUB-STRAND 3.1: RADIOACTIVITY AND STABILITY OF ISOTOPES	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

Your Final Explanation is your chance to show everything you have learned across all 7 lessons about radioactivity. You will write a structured, evidence-based response to the driving question below. Read the driving question carefully before you begin.

DRIVING QUESTION: "Why do some atoms in Kenyan rocks spontaneously release invisible energy — and how can we harness or protect ourselves from that energy?"

BEFORE YOU WRITE:

- Re-read your Driving Question Board (DQB) entries and any notes you made during the lessons.
- Think back to the anchoring phenomenon — the pitchblende ore that exposed photographic film in complete darkness. Every section of your answer should connect back to explaining what is really happening inside that rock, and why it matters.
- Use scientific vocabulary correctly. Key terms you should use include: nucleus, proton, neutron, isotope, nucleon number, proton number, belt of stability, radioactive decay, alpha (α) particle, beta (β) particle, gamma (γ) radiation, nuclear equation, half-life, ionisation, Geiger-Müller tube, and fission/fusion.
- Use Kenya-relevant examples wherever possible — Kenyan geology, hospitals, agriculture, and energy contexts are all fair game.
- Write in full sentences and organised paragraphs. Each section has a guiding prompt and a word-count guide to help you.

ASSESSMENT INFORMATION:
This Final Explanation is worth a major portion of your Sub-strand 3.1 grade. Your teacher will mark your response using the rubric at the end of this document. Aim for the "Excellent" descriptor in every criterion. There is no single "right" answer — what matters is that your explanation is accurate, well-reasoned, connected to evidence, and clearly communicated.

Section 1 — The Mystery of the Rock: Nuclear Structure and Isotopes	
Explain why the pitchblende rock was able to expose the photographic film without any	When the pitchblende ore was placed next to photographic film in complete darkness, it still caused an exposure — not because of light or heat, but because of radiation released spontaneously from inside the atoms of the rock itself. To understand why, we need to look at the nucleus. Every atom has a nucleus made up of protons and neutrons, collectively called nucleons. The proton

light, heat, or sound. In your answer, describe the structure of an atom's nucleus (protons, neutrons, nucleon number, proton number) and explain what an isotope is. Use uranium as your main example. Why do different isotopes of the same element behave so differently? (Suggested length: 150–200 words)	number (Z) tells you how many protons are present and determines which element the atom is. The nucleon number (A) is the total count of protons and neutrons. Isotopes are atoms of the same element that share the same proton number but have different numbers of neutrons — and therefore different nucleon numbers. Uranium, the main radioactive element in pitchblende, has several isotopes. Uranium-238 (${}^{238}_{92}U$) and Uranium-235 (${}^{235}_{92}U$) both have 92 protons, but U-238 has 146 neutrons while U-235 has only 143. This difference in neutron count dramatically changes how stable the nucleus is. Stable isotopes hold together indefinitely; unstable isotopes — like those found in pitchblende ore — spontaneously break apart, releasing energy in the process. That released energy is what darkened the film.

Section 2 — Why Some Nuclei Are Unstable: The Belt of Stability and Radioactive Decay	
Explain what the 'belt of stability' is and how it helps us predict whether a nucleus is likely to be radioactive. Then describe the three main types of radioactive decay (alpha, beta, and gamma), including what is emitted in each, how far each type travels, and what each type can penetrate. Write and explain at least ONE balanced nuclear equation for a decay process. Connect your answer back to what is happening inside the pitchblende rock. (Suggested length: 200–250 words)	Not all nuclei are equally stable. Scientists have mapped the neutron-to-proton (n/p) ratio of every known isotope on a graph called the belt of stability (or valley of stability). Nuclei that fall within this belt are stable; those that fall outside it are unstable and will undergo radioactive decay — spontaneously reorganising their nuclei to move closer to stability. Uranium-238, found in the pitchblende rock, lies far outside the belt, which is exactly why it keeps emitting radiation without any external energy input. There are three main types of radioactive decay. Alpha (α) decay occurs when the nucleus ejects an alpha particle — a cluster of 2 protons and 2 neutrons, identical to a helium-4 nucleus. Alpha particles travel only a few centimetres in air and are stopped by a sheet of paper or skin, but they are highly ionising. For example, the alpha decay of Uranium-238 can be written as: ${}^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$. Notice that the nucleon number decreases by 4 and the proton number decreases by 2 — the equation is balanced on both sides. Beta (β) decay occurs when a neutron in the nucleus converts into a proton and a high-speed electron (the beta particle) is ejected. Beta particles can travel about a metre in air and are stopped by a few millimetres of aluminium. Gamma (γ) radiation is not a particle but a high-energy electromagnetic wave released when the nucleus loses excess energy after a decay. It is the most penetrating, requiring several centimetres of lead or metres of concrete to be significantly absorbed. Inside the pitchblende rock, all three types of decay are happening across a long chain of unstable nuclei, which is why the rock never seems to 'run out' of radiation on a human timescale.

Section 3 — Measuring the Invisible: Detection and Half-Life	
Describe how scientists and medical professionals detect radiation (include at least TWO detection methods, e.g. Geiger-Müller tube, photographic film, cloud chamber). Then explain the concept of half-life: what it means, how it is calculated or read from a decay curve, and	Because radiation is invisible, scientists use special instruments to detect it. The photographic film in the original pitchblende experiment was itself a detector — ionising radiation exposes silver halide crystals on the film just as light would. A more precise instrument is the Geiger-Müller (GM) tube, which contains a gas that becomes ionised when radiation passes through it. Each ionisation event creates a small electrical pulse that is counted and displayed as 'counts per second' or 'counts per minute.' Medical workers and researchers in Kenya — such as those at Kenyatta National Hospital's nuclear medicine unit — use personal dosimeter badges (a type of photographic film detector) to monitor their cumulative radiation exposure over weeks. Half-life is the time taken for half of the radioactive nuclei in a sample to decay. It is a constant for each isotope. For example, if a sample contains 800 g of a radioactive isotope with a half-life of 10 years, then after 10 years only 400 g remains undecayed; after

why it is useful. Give a worked example using realistic numbers. How does half-life explain why the pitchblende rock has been emitting radiation for millions of years? (Suggested length: 150–200 words)	20 years, 200 g; after 30 years, 100 g; and so on. You can read half-life directly from a decay curve by finding the time at which the count rate (or mass) has dropped to exactly half its starting value. Uranium-238 has an extraordinarily long half-life of approximately 4.5 billion years — almost the same age as Earth itself. This is why the pitchblende rock has been continuously emitting radiation for millions of years and will continue to do so long into the future: the vast majority of its U-238 nuclei have not yet decayed.

Section 4 — Harnessing the Energy: Beneficial Applications in Kenya	
Describe at least THREE specific ways in which knowledge of radioactivity benefits people in Kenya. You must include examples from at least TWO different fields (e.g. medicine, agriculture, energy, environmental monitoring, or archaeology). For each application, explain the underlying nuclear science that makes it work. (Suggested length: 150–200 words)	Understanding radioactivity has unlocked powerful applications that directly improve life in Kenya across several fields. In medicine, radioactive isotopes are used as tracers and in cancer treatment. At Kenyatta National Hospital, radioactive iodine-131 (I-131) is used to treat thyroid cancer. Because the thyroid gland naturally absorbs iodine, the radioactive version accumulates there and its beta and gamma emissions destroy cancerous cells from within. This works precisely because of the ionising properties of radiation that we observed throughout this sub-strand. In agriculture, the Kenya Agricultural and Livestock Research Organisation (KALRO) uses radiation-induced mutation techniques to develop improved crop varieties. Seeds — for example, maize or sorghum — are exposed to carefully controlled doses of gamma radiation, which causes mutations in the DNA. Scientists then select plants with desirable traits such as drought resistance, a critical issue in Kenya's semi-arid regions like Kitui and Garissa. In energy, nuclear fission — the splitting of heavy nuclei such as uranium-235 — releases enormous amounts of energy. Although Kenya currently leads in geothermal and solar energy, understanding fission is essential as Africa considers nuclear energy to meet growing electricity demands. Fission contrasts with fusion (joining of light nuclei), which powers the Sun and represents the long-term future of clean energy globally.

Section 5 — Protecting Ourselves: Radiation Hazards and Safety	
Explain why radiation is hazardous to living things at the cellular and molecular level. Then describe at least THREE practical safety measures that should be used when working with radioactive materials. Finally, return to the anchoring phenomenon one last time: now that you have completed all 7 lessons, write 3–4 sentences that fully answer the driving question in your own words,	Radiation is dangerous to living things because it is ionising — it carries enough energy to knock electrons out of atoms inside body cells. When this happens in DNA molecules, it can break chemical bonds, cause mutations, or prevent cells from dividing correctly. High doses can cause radiation sickness, while chronic low-level exposure increases the risk of cancers such as leukaemia. This is why the pitchblende rock, left unattended in a room for a long period, would pose a genuine health risk to anyone nearby. Three essential safety measures are: (1) Distance — the intensity of radiation decreases with the square of the distance from the source, so moving further away significantly reduces exposure; workers use long-handled tongs to handle radioactive samples rather than touching them directly. (2) Shielding — placing dense materials between the source and the body absorbs radiation; lead aprons are used in X-ray rooms at Mombasa Coast General Hospital, for instance. (3) Time — minimising the duration of exposure reduces the total dose received; radiation workers rotate shifts to limit individual exposure.

connecting the pitchblende rock all the way through to human health, Kenyan applications, and nuclear science. (Suggested length: 150–200 words)	In summary: the pitchblende rock exposes film because uranium-238 nuclei — whose neutron-to-proton ratio places them far outside the belt of stability — spontaneously decay, releasing alpha, beta, and gamma radiation. This process occurs without any external energy input because the nucleus itself reorganises to reach a more stable state. By understanding this nuclear science, Kenyans can harness radiation in hospitals, farms, and future energy systems, while using shielding, distance, and careful monitoring to stay safe.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Nuclear Science Accuracy	All scientific concepts (nuclear structure, isotopes, belt of stability, decay types, nuclear equations, half-life, fission/fusion) are explained correctly and with precise use of terminology. Nuclear equations are balanced correctly with nucleon and proton numbers conserved. No factual errors are present.	Most scientific concepts are explained correctly with mostly accurate terminology. Nuclear equations are attempted and are mostly balanced. Minor errors or imprecision present but do not undermine the overall explanation.	Some scientific concepts are present but contain significant inaccuracies or misconceptions. Terminology is used incorrectly or inconsistently. Nuclear equations are missing, unbalanced, or contain major errors.
Connection to Anchoring Phenomenon	Every section explicitly and meaningfully connects back to the pitchblende rock phenomenon. The final summary clearly and completely answers the driving question by integrating all key concepts (nuclear instability → decay types → detection/half-life → applications → safety) into a coherent narrative.	Most sections connect to the pitchblende phenomenon. The driving question is addressed in the final section but the answer may lack full integration of all key concepts or may leave some threads unexplained.	Connections to the pitchblende phenomenon are superficial, rare, or absent. The driving question is not clearly answered, or the answer addresses only one or two aspects of the question.
Kenyan and Real-World Application	At least three specific, correctly explained Kenyan applications are given across at least two fields (medicine, agriculture, energy, environmental monitoring). Kenyan institutions, place names, or scientists are referenced accurately and add genuine context rather than being decorative.	At least two Kenyan applications are given across one or two fields and are mostly correctly explained. Kenyan context is present but may be generic or partially accurate.	Fewer than two applications are given, or applications are described without explaining the underlying nuclear science. Little or no Kenyan context is present.
Use of Evidence and Scientific Reasoning	Claims are consistently supported with specific scientific evidence or reasoning (e.g. specific isotopes, numerical half-life values, decay equations, penetration data). The student demonstrates understanding of cause-and-effect relationships at the nuclear level and clearly distinguishes between	Most claims are supported with some evidence or reasoning. The student shows understanding of most cause-and-effect relationships but may occasionally state facts without explanation or make logical leaps without support.	Claims are frequently unsupported or based on recall without reasoning. Cause-and-effect relationships at the nuclear level are unclear or absent. The response reads as a list of facts rather than a reasoned explanation.

	observation and explanation.		
Communication and Structure	The response is clearly organised into all five sections with well-developed paragraphs. Scientific vocabulary is used correctly and consistently throughout. Writing is fluent, precise, and accessible. Spelling and grammar are accurate. The response meets the suggested word-count guidance for each section.	The response addresses all five sections with adequate organisation. Scientific vocabulary is mostly used correctly. Writing is generally clear with occasional lapses in fluency or precision. Minor spelling/grammar errors that do not impede understanding.	The response is poorly organised, incomplete (one or more sections missing or very brief), or difficult to follow. Scientific vocabulary is limited or frequently misused. Frequent spelling/grammar errors impede understanding.